Class	Adm No

Candidate Name: ______________________________

H

millennia institute

2014 Preliminary Examination II
Pre-university 3

H2 Biology **9648/03**

Paper 3 **22 September 2014**

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.
The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
Total	

For Examiner's Use

Answer **all** questions

1. The screwworm fly, *Cochliomyia hominivorax*, is a pest of livestock. The size of its population can be reduced by releasing large numbers of sterile male flies into the environment. The sterile male flies mate with wild female flies, but no offspring are produced.

Large-scale release of genetically modified (GM) sterile males in the 1950s took place in southern USA. Each GM fly contained a gene coding for green fluorescent protein (GFP) taken from a species of jellyfish, used as a genetic marker. The added DNA also included a promoter.

(a) Explain the purpose of using a genetic marker in genetic engineering.

- To allow identification of recombinants;
- Genetic marker provides an observable phenotype.

[2]

(b) Explain why genes for fluorescent proteins such as GFP are now commonly used as genetic markers instead of genes for antibiotic resistance.

- Antibiotic resistance genes have risk of spreading to bacteria in the wild via horizontal gene transfer;
- Other strains of bacteria can develop resistance to antibiotics;
- Phenotype of fluorescent protein genes can be easily visualized.

[2]

(c) Explain why a promoter needs to be included when transferring a gene from jellyfish into the screwworm fly.

- Promoter is required for RNA polymerase to bind;
- To ensure that gene is transcribed;
- Gene may be inserted into a region of the genome where there is no promoter.

[2]

Although GFP is visible in the screwworm fly, allowing detection of GM flies, confirmation of the presence of the newly added gene can be done via genetic fingerprinting. The first stage of genetic fingerprinting involves using restriction enzymes to digest fly genomic DNA and conducting gel electrophoresis.

(d) Explain how the presence of the GFP gene can be detected after gel electrophoresis is complete.

- Transfer DNA from gel to nitrocellulose membrane using capillary action;
- Expose nitrocellulose membrane to radioactively labelled probe with sequence complementary to GFP gene;
- Use autoradiography to detect bands that radioactive probe has bound to.

.... [3]

Fig. 1.1 shows the restriction map of the plasmid that was used to clone the sterility gene into male screwworm flies.

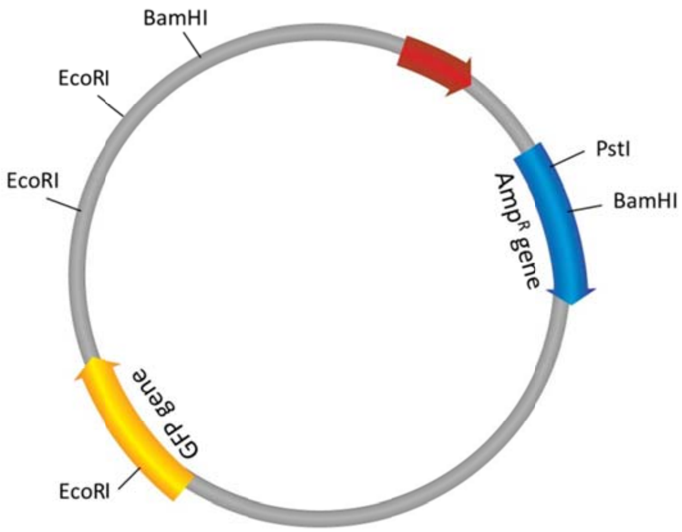

Fig. 1.1

(e) Suggest and explain which restriction enzyme should be used to cleave the plasmid for insertion of the sterility gene.

- PstI
- PstI only has one restriction site on the plasmid that is located within a genetic marker gene;
- Pst1 will not produce multiple restriction fragments.

.... [2]

Before the gene for sterility can be cloned into the plasmid, scientists have to generate many copies of the gene using the Polymerase Chain Reaction (PCR). Fig. 1.2 shows the changes in temperature at different stages of the PCR process.

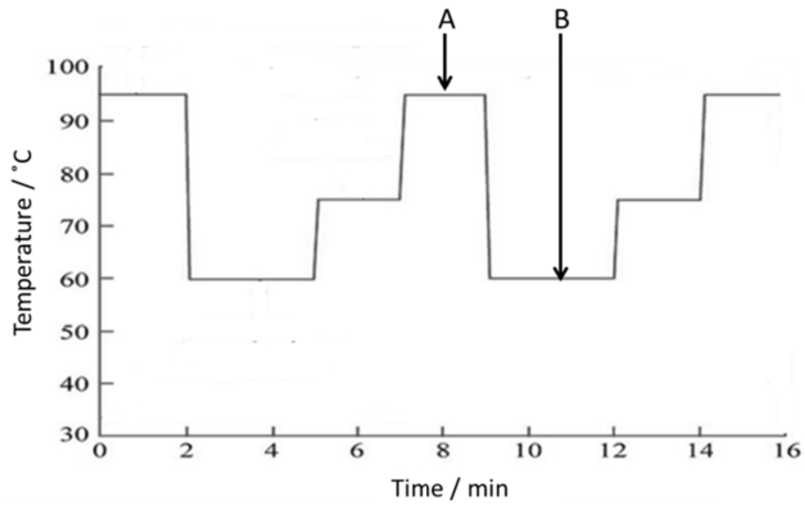

Fig. 1.2

(f) With reference to Fig. 1.2, explain the purpose of stages A and B.

- Stage A: Denaturation
- Heat to 94˚C to separate double-stranded DNA by <u>disrupting hydrogen bonds</u>;
- Stage B: Annealing
- Lower to 60˚C to allow annealing of primers to 3' end of template strands via <u>complementary base pairing</u>.

[4]

(g) When one scientist conducts PCR using the conditions shown in Fig. 1.2, he finds that the final mixture contains DNA fragments of varying lengths, instead of only one length.

Suggest why the final mixture contains DNA fragments of varying lengths.

- Non-specific binding of primers to template strands.

[1]

[Total: 16]

2. Severe combined immunodeficiency (SCID) is a genetic disorder characterised by the absence of functional T-lymphocytes. There are two common forms of SCID, namely X-linked SCID and adenosine deaminase (ADA) deficiency SCID.

(a) Explain the difference in cause of X-linked SCID and ADA deficiency SCID.

- X-linked SCID is due to mutations in gene coding for common gamma chain in interleukin receptors;
- ADA deficiency SCID is due to mutated gene coding for adenosine deaminase.

.... [2]

One method of treating ADA-deficiency SCID is through the use of gene therapy, where stem cells are isolated and the non-functional ADA gene is replaced with the functional gene. The optimal number of functioning white blood cells is approximately 5000 – 8000 per mm^3 of blood.

Two patients undergo the same stem cell gene therapy for ADA-deficiency SCID, using stem cells isolated from their own bodies. Retroviral vectors were used to transport the functional ADA gene into isolated stem cells. The results of the therapies are shown in Fig. 2.1.

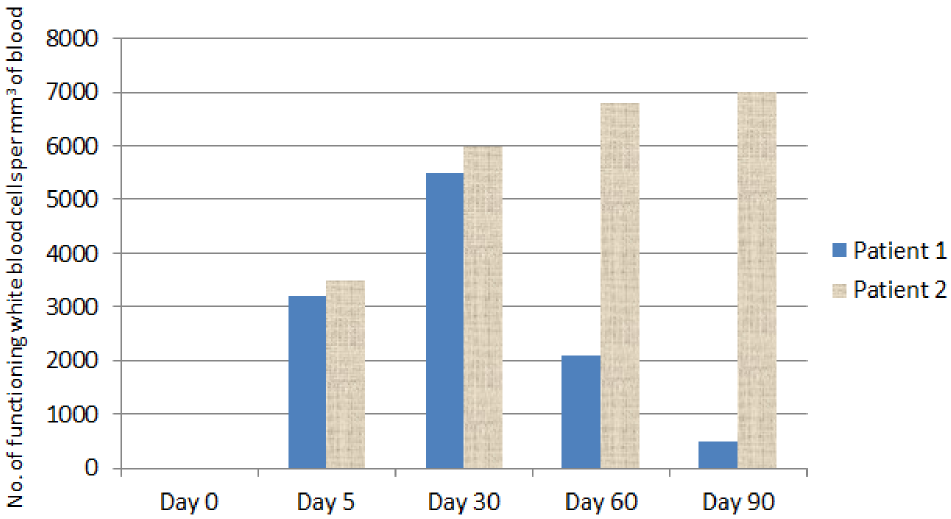

Fig. 2.1

(b) Suggest and explain what type of stem cells could be used for treatment of SCID.

- Haematopoietic stem cells; R adult stem cells / multipotent stem cells
- They can differentiate to form different types of blood cells, including white blood cells.

.... [2]

(c)

(i) With reference to Fig. 2.1, describe the differences between the effects of gene therapy in patient 1 and patient 2.

- For patient 1, no. of functioning white blood cells increases from day 0 to day 30, from 0 to ~6000 cells per mm^3 but decreases again to ~500 cells per mm^3 by day 90;
- For patient 2, no. of functioning white blood cells increases steadily from day 0 to day 60, from 0 to ~6800 cells per mm^3 of blood, starting to plateau at day 90.

[2]

(ii) Suggest an explanation for the difference in the results of treatment for patients 1 and 2.

- Treatment was a success for patient 1 as stem cells were able to differentiate into functional white blood cells containing ADA gene;
- Treatment was not successful for patient 2 possibly because retroviral vector regained virulence and began to destroy treated cells.

[2]

(d) Explain why doctors prefer to use stem cells for gene therapy as compared to somatic cells.

- Stem cells have self-renewal capability thus there is no need for repeated treatments;
- Stem cells can differentiate into a variety of cell types including white blood cells/lymphocytes that carry the normal functional allele;
- When somatic cells die, therapeutic effect is lost as new cells do not contained normal allele.

[2]

(e) Suggest two advantages of using viruses to transport the ADA gene into stem cells.

- Specificity of target cells due to recognition of specific cell surface receptors by viral glycoproteins;
- High transfection efficiency at suitable concentration;
- Some viruses are able to integrate gene into host cell genome, which stabilises gene expression.

Any 2

[2]

[Total: 12]

3. Duchenne muscular dystrophy (DMD) is a genetically inherited disorder involving a deficiency in the dystrophin protein. Individuals with the disease experience progressive muscle weakness and eventual death. DMD is more common in males than females.

Fig. 3.1 shows a pedigree chart for a family with inheritance of DMD and the restriction pattern of different members of the family obtained by detection of an RFLP closely linked with the DMD gene.

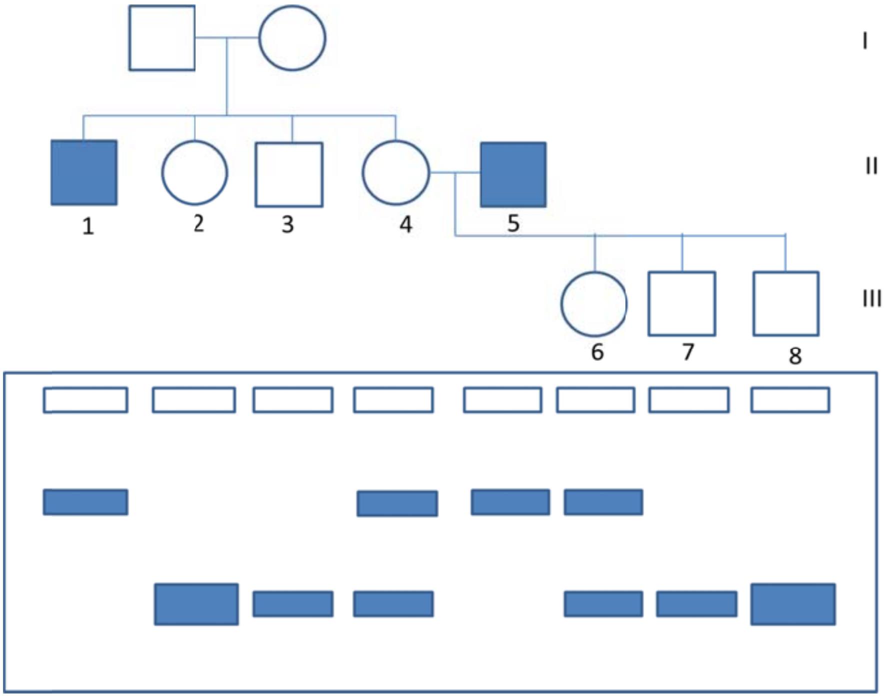

Fig. 3.1

(a)

(i) With reference to Fig. 3.1, state the mode of inheritance of DMD.

- Sex-linked / X-linked

.... [1]

(ii) Based on the restriction pattern shown in Fig. 3.1, deduce the genotype of individual III-6.

- X^RX^r; R heterozygous
- Individual III-6 exhibits 2 restriction bands, one representing the dominant allele and the other representing the recessive allele.

.... [2]

For Examiner's Use

(iii) Suggest why there is a difference in band thickness between individuals II-2 and II-4 even though they exhibit the same phenotype.

- II-2 is homozygous dominant whereas II-4 is heterozygous;
- II-2 has two copies of the dominant allele whereas II-4 only as a single copy of the allele.

[2]

(iv) One of the individuals represented in Fig. 3.1 exhibits an anomalous restriction pattern.

Identify which individual exhibits the anomalous pattern and suggest a reason for the anomaly.

- Individual III-8 has a thick band which indicates that he should have two copies of the dominant allele;
- However, III-8 is a male and should only have one X chromosome;
- Non-disjunction could have occurred resulting in him having genotype XXY / having two X chromosomes and one Y chromosome.

[3]

In disease detection, RFLPs are often used as genetic markers to predict whether or not a person is likely to suffer from a particular disease.

(d) Explain the features that must be present in order for a genetic marker to be used as a means of determining inheritance of a disease.

- Marker must be polymorphic so different form of marker is associated with different allele;
- Marker must be closely linked to disease gene so that they will be inherited together / not affected by crossing over.

[2]

(e) Suggest two other uses for RFLPs other than for disease detection.

- To create linkage map;
- DNA fingerprinting to identify individuals;
- Paternity testing;

AVP

[2]

[Total: 12]

4. Planning Question

Bacteria have the ability to acquire new genes via horizontal gene transfer. One of the methods of horizontal gene transfer is the process of transformation, where bacteria are able to take up naked DNA from the surroundings. The process of transformation has been utilised by scientists to introduce foreign DNA into bacteria cells for gene cloning purposes. This is often done using the heat-shock method.

You are required to plan, but not carry out, an investigation into the effect of heat-shock temperature on the efficiency of bacterial transformation, using a range of temperatures between 30˚C and 80˚C.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you must use:

- Solution of plasmid containing tetracycline resistance gene
- Suspension of *E.coli* cells
- Distilled water
- LB nutrient broth
- Petri dishes containing LB agar medium and tetracycline antibiotic
- Calcium chloride solution
- Micropipettes
- Microfuge tubes
- Stopwatch
- Sterile streaking loops
- Parafilm
- Ice
- Access to hot water at 100˚C
- Access to incubator at 37˚C
- Alcohol disinfectant

Your plan should have a clear and helpful structure to include:

- an explanation of theory to support your practical procedure
- a description of the method used including the scientific reasoning behind the method
- an explanation of the dependent and independent variables involved
- relevant, clearly labelled diagrams
- how you will record your results and ensure they are as accurate and reliable as possible
- proposed layout of results, tables and graphs with clear headings and labels
- the correct use of technical and scientific terms

[Total: 12]

Theoretical consideration / rationale to justify experimental procedure [Max 4]

- Bacteria cells have to be made competent using calcium chloride before conducting heat-shock;
- Pores temporarily appear in membrane of bacteria during heat-shock to allow uptake of DNA;
- Increase in temperature will increase kinetic energy of membrane phospholipids, increasing the rate of pore formation and DNA uptake.
- Plasmid with tetracycline resistance gene is used to identify bacteria that have successfully taken up the plasmid;
- Recombinant bacteria will be able to successfully grow on LB/tet medium;
- Increased number of recombinant bacteria colonies should be observed with increasing heat-shock temperature;
- Above optimum temperature, transformation efficiency will decrease as cell membrane is disrupted and cells die.

Procedure [Max 4; 2 points awarded 1 mark]

- Independent variable: heat-shock temperature – at least 5 specific temperatures given at equal intervals between 30-80˚C
- Dependent variable: number of colonies growing on LB/tet plates
- Mix fixed volume of *E. coli* culture with fixed volume of calcium chloride – to make cells competent;
- Add fixed volume of plasmid solution to *E. coli* culture;
- Heat-shock procedure: Incubate <u>E. coli</u> on ice for 1 minute [fixed amount of time no longer than 2 minutes]; Transfer *E. coli* to water bath at 30˚C for 90 seconds; Back on ice.
- Plate *E. coli* on LB/tet agar plate using sterile inoculating loops and seal with parafilm;
- Incubate plates overnight in 37˚C incubator and count number of colonies on plate;
- Conduct 3 replicates for each heat-shock temperature and repeat the entire experiment once to ensure accuracy and reliability.

Control [1]

- Replaced plasmid solution with <u>equal volume</u> of distilled water to show that survival of bacteria on LB/tet plate is due to presence of plasmid.

Results and Data [2]

- Table with appropriate headings and units

Heat-shock temperature / ˚C	No. of colonies			
	Plate 1	Plate 2	Plate 3	Average
30				
40				
50				
60				

- Graph with appropriate axes labels and units showing correct trend;

Appropriate use of scientific and technical terms [1]

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5. Free-response question

Write your answer to the question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate,
- must be in continuous prose, where appropriate,
- must be set out in sections (a), (b) etc, as indicated in the question.

(a) Both traditional vegetative propagation techniques like grafting and tissue culture techniques can be used to clone crop plants.

Explain the advantages of tissue culture techniques over traditional vegetative propagation techniques. [5]

- Easier to produce hybrids to obtain new varieties of plants;
- Enables preservation of pollen and cell collections from which plants may be propagated;
- Decreasing amount of space required for field trials;
- Selection of desirable traits is possible directly from culturing set-up;
- Genetic engineering of male sterility can be carried out, which would increase efficiency of conventional breeding programmes by preventing self-pollination;
- Reduces plant disease through careful selection of stock plants and use of sterile techniques;
- Enables cold storage of large numbers of viable plants in small space.

Max 5

(b) Bovine somatotrophin (BST) is a peptide hormone that is produced by the pituitary glands of cows. Administration of BST to cows is able to increase the milk production rate of cows, thus increasing farmers' yields.

Using your knowledge of genetic engineering, explain how BST could be produced in large quantities using bacteria. [10]

- Isolate BST mRNA from cow genome using restriction enzymes;
- Use reverse transcriptase to convert BST mRNA to cDNA;
- Use same restriction enzymes to cleave BST cDNA and plasmid;
- BST gene should be inserted within marker gene e.g. ampicillin resistance gene;
- Use DNA ligase to catalyse insertion of BST gene into plasmid vector;
- Introduce recombinant plasmid into bacteria cells via transformation E.g. of transformation method – heat-shock / electroporation etc;
- Conduct replica plating to identify bacteria that contain recombinant plasmid;
- Transfer colonies from replica plate to nitrocellulose membrane – Southern blotting;
- Use NaOH to burst bacteria cells and denature dsDNA into ssDNA;
- Wash membrane with chloroform and saline and bake in oven to fix DNA onto filter;

- Add radioactive gene probe with sequence complementary to BST gene / nucleic acid hybridsation;
- Locate position of radioactive gene probe using autoradiography;
- Isolate bacteria colonies containing BST gene and grow in large quantities using bioreactor.

Max 10

(c) Despite the advantages offered by genetically modified (GM) crops, the European Union exercises highly stringent regulations with regards to GM crops and only 48 GM crops have been approved so far. Most of these crops are only approved as animal feed and not for human consumption.

Suggest why the use of GM crops is not widespread despite its advantages. [5]

- Antibiotic resistance genes used in genetic engineering may be passed on to harmful bacteria in the surroundings;
- GM crops may establish themselves as weeds as they have selectively advantageous characteristics/can outcompete other plants;
- Potential spread of resistance from GM crops to weeds;
- Potential upset of ecosystem by introducing GM organisms into environment;
- Foreign genes may be introduced to other organisms via bacterial and viral vectors;
- Possibility of allergic reactions due to inaccurate labelling – GM crops contain genes from other species;

Max 5

[Total: 20]

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